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Feasibility study and performance analysis of microgrid with 100% hybrid renewables for a real agricultural irrigation application

Mahmoud F. Elmorshedy [a,b], Mohamed R. Elkadeem [a,*], Kotb M. Kotb [a,*], Ibrahim B.M. Taha [c], Mohamed K. El-Nemr [a], A.W. Kandeal [d], Swellam W. Sharshir [d], Dhafer J. Almakhles [b], Sherif M. Imam [e]

[a] Electrical Power and Machines Engineering Department, Faculty of Engineering, Tanta University, Tanta 31521, Egypt
[b] Renewable Energy Lab, College of Engineering, Prince Sultan University, Riyadh, Saudi Arabia
[c] Electrical Engineering Dept., College of Engineering, Taif University, P. O. Box. 11099, Taif 21944, Saudi Arabia
[d] Mechanical Engineering Department, Faculty of Engineering, Kafrelsheikh University, Kafrelsheikh 33516, Egypt
[e] Electrical Power Engineering Department, Faculty of Engineering, Damanhour University, 22511 Damanhour, Egypt

A R T I C L E I N F O

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A B S T R A C T

Recently, wide attention has been paid to the techno-economic viability for electrifying rural and remote areas while less awareness for agricultural irrigation pumps is given. Diesel generators are still utilized for agricultural facilities, causing ecological issues and energy price growth. In this paper, an integrative techno-economic design optimization framework for adequate planning of fully renewable energy system including photovoltaic units, wind turbines and batteries fed a real case-study of an agricultural load of 240 kWh/day in Al-Sadat city, Egypt, is proposed using HOMER Pro. The project site is favorably chosen since Egypt holds great potential for renewables; manipulating these resources becomes crucial for achieving national sustainability as per Egypt 2030 vision. The feasibility and performance of the proposed energy system are evaluated and analyzed regarding its technical and economic dimensions and compared with four different energy alternatives, including the existing real system and the standalone diesel system, for fair comparisons. The design optimization is followed by parametric and reliability analyses to uncover the impacts of uncertainty in seven key input parameters and power generation shortage on optimal economic behavior. The achieved outcomes reveal that the fully renewable solar/wind/batteries/converter system is of optimal behaviours over other alternatives to fulfil the load demand. The proposed system acquires the minimum net present cost ($194,017) and cost of energy (0.119 $/kWh) and efficiently attains a negligible ratio in both capacity shortage and unserved load with only 0.098%. Besides, a productive amount of surplus energy (61.7%) is achieved for further deferable loads supplying. Moreover, the parametric analysis proves the great reliance of system cost on the uncertainty of discount rate, project lifetime and components' capital costs while the uncertainty in renewable resources has a slight impact on the system performance. In addition, the reliability analysis indicates that a small and tolerable loss of power supply by 2% results in considerable financial benefits while optimizing the microgrid design. The study is expected to raise the awareness of exploiting renewable energy in Egypt and provide a valuable proposal for replacing the existing energy system of the addressed case-study.

Abbreviations: BAT, Batteries; BESA, Bald eagle search algorithm; CO2, Carbon dioxide; COE, Cost of energy; CONV, Converter; CSF, Capacity shortage fraction; DG, Diesel generator; DOD, Depth of discharge; GA, Genetic Algorithm; HOMER, Hybrid Optimization Model for Electric Renewables; HT, Hydrogen tank; LF, Load following; LLP, Load loss probability; LS, Load shifting; MG, Microgrid; NPC, Net present cost; PV, Photovoltaic; RF, Renewable friction; SE, Surplus energy; SOC, State of charge; STC, Standard test conditions; TCE, Total carbon emission; UL, Unmet load; VCS, Virus colony search; WCA, Water cycle algorithm; WTs, Wind turbines.

* Corresponding authors.
E-mail addresses: mahmoud.elmorshedy@f-eng.tanta.edu.eg (M.F. Elmorshedy), mohammad.elkadim@f-eng.tanta.edu.eg (M.R. Elkadeem), kotb.mohamed@f-eng.tanta.edu.eg (K.M. Kotb).

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Nomenclatures

Photovoltaic

$P_{PV,r}$	rated power of the PV module
f_{PV}	Derating factor
G_T	Average hourly solar radiation at nominal working temperature
$G_{T,\ STC}$	Solar radiation under the Standard Test Conditions
β_r	Temperature coefficient of power or the efficiency of SPV cell (%/°C)
T_{PV}	Actual temperature of the SPV cell (°C)
$T_{PV,\ STC}$	Temperature of the SPV cell at STC
γ	SR intensity coefficient for the solar cell efficiency
$\eta_{T,\ STC}$	PV module efficiency measured at STC
T_{amb}	Ambient temperature
T_{NOCT}	Nominal operating cell temperature

Diesel Generator

f_{gr}	DG fuel curve intercept coefficient
P_r	Rated capacity of the utilized DG
f_{go}	DG fuel curve slope

Battery storage

E_L	Desired daily load energy consumption
γ_{DA}	Daily autonomy
η_{CONV}	Converter efficiency
DOD	Battery depth of discharge
P_L	Load demand at time t
η_{BAT}	Round-trip efficiency of the battery

System converter

$\eta_{DC/AC}$	Converter conversion efficiencies from DC to AC
P_{CONV}	Converter capacity in kW
$\eta_{AC/DC}$	Converter energy conversion efficiencies from AC to DC

Wind Turbine

$P_{WT,\ r}$	Rated output power of the wind turbine
$V_{\omega,h}$	Wind speed at the turbine's hub height
V_{anem}	Wind speed at anemometer height
h_{ref}	Surface roughness length
h_{anem}	Anemometer height
h_{hub}	Hub height of the turbine
ρ_{act}	Air density
ρ_{ref}	Standard Temperature and Pressure
V_{in}	Cut-in
V_{off}	Cut-off wind
V_r	Turbine's rated speed
P_g	Generated power of DG
ρ_f	Density of fuel
H_{lhf}	Lower heating value of the diesel fuel
α	Battery self-discharge rate per hour
Δt	Time interval (1hr).
$E_{BAT}(t)$	Storage energy of the battery at time t
E_{BAT} (t-1)	Storage energy of the battery at time t-1
P_g	Microgrid generated power

Introduction

Background

The global population growth and large use of fossil fuels-based generators have caused many greenhouse gases, mainly in the form of CO_2 emissions, and led to tremendous environmental harm [1]. In the global breakdown of emissions by sector, agriculture is the fourth biggest source of CO_2 with 12.68 % [2]. Also, over 70 % of freshwater is withdrawn globally for agriculture to keep fruits, vegetables, and grains growing for food security and improve the global supply chain [3]. Thus, the agriculture sector provides opportunities and prompts to integrate renewable energy (RE) sources, mitigate climate change and manage irrigation water demand [4]. Today, microgrids (MGs) offer a viable solution for integrating distributed energy resources, including in particular variable and unpredictable renewable energy sources (i.e., photovoltaic (PV), wind turbines (WTs), low-voltage and medium-voltage into distribution networks [5]. MG refers to the integration of many renewable and conventional generation sources and energy storage systems to supply different load demands. It is considered the most suitable way to merge all these technologies in a single reliable platform [6]. The emergence of autonomous MG technology fully plays the value and benefits of RE sources [7]. To have a cost-effective and sustainable RE-based microgrid, optimal sizing is necessary. However, it is a complex task due to the many variables and constraints that depend on the available power supply components, local resources data, technical and cost specifications, and load profiles [8]. The undersizing can lead to operation failure and largely unmet demand. Meanwhile, the reliability problem is solved by oversizing; however, it may result in high system costs with excessive energy generated [9].

Literature survey

In the past 10 years, numerous studies have been offered to analyze, appraise, and review the optimal design and feasibility analysis of the different structures of MGs for supplying different types of loads over the global [10,11,8,12]. For example, Rinaldi et al. [13] optimized a hybrid PV/ WT/diesel generator (DG)/batteries (BAT)/system converter (CONV) microgrid for off-grid electrification of three different remote regions with different climates in Peru. The optimal microgrid sizing (with minimum net present cost "NPC") was conducted via HOMER software regarding seven scenarios (separate and hybrid components) and considering climate, load demand, fuel price, and components' cost. It was concluded that the (PV/WT/DG) system was the most cost-effective with NPC, cost of energy (COE), RF ranges of about ($0.1466–$0.2273 million), (0.460–0.504$/kWh), and (94–97 %), respectively. In addition, the CO_2 emissions ranged between 2.7 and 9.9 % of that of separate DG. Salisu et al. in [14] obtained four optimum microgrid scenarios of energy systems for Nigerian rural villages via HOMER: PV/DG/BAT, PV/WT/DG/BAT, PV/WT/BAT, and PV/BAT. As concluded, the optimal design was (PV/DG/BAT) with 0.110$/kWh (COE), $1.01 million (NPC), 98.3 % (RF), and 2.89 ton/yr. (CO_2 emission). The COE and NPC varied directly with the irradiance and inversely with the fuel price. Similarly, Berbaoui et al. [15] sized and optimized PV/DG/BAT system for a remote Algerian area using Virus Colony Search (VCS) algorithm. The results revealed that PV/DG/BAT is the most economic scenario, which became more cost-effective in the long-term when multiple DG were incorporated. Whereas, in the case of a single DG, the PV/DG was more cost-effective than PV/DG/BAT because the BAT could not significantly lessen the DG required continuous operation. Tsai et al. [16] assessed the techno-economic feasibility of PV/DG/BAT/CONV microgrid proposed for Taiwanese island simulated via HOMER and compared different scenarios. As reported, the hybrid PV/DG system had the lowest COE (0.3569$/kWh) with a PV system capacity of 200 kWp, and RF of 15.3 %. Besides, the standalone DG had lower COE than PV/DG/BAT/CONV, whereas the latter reduced the CO_2 emissions by 31.63 %.

Moreover, Eras-Almeida et al. [17] utilized HOMER Pro software to conduct a techno-economic analysis of a hybrid renewable mini-grid (PV/WT/DG/BAT/COV) considering various energy demands of Galapagos islands. As summarised, the addition of PV (18.25 MWp) and BAT (20.68 MWh) could reduce the COE by 40.9 % and increase the RF by 116.6 %. Besides, more reduction in COE could be obtained by applying the energy efficiency measures. Liu et al. [18] techno-economically assessed different microgrid structures to stratify high-rise residential buildings' demand. The analysis was conducted via modelling (TRNSYS) and optimization (jEPlus + EA). The optimal system (PV/WT/BAT) had annual demand's coverage ratio and COE of (81.29 % and 0.2230 \$/kWh) compared to (16.02 % and 0.5252 \$/kWh), and (53.65 % and 0.1251 \$/kWh) for PV, and PV/WT systems, respectively. Alsagri et al. [19] considered a PV/DG/BAT/CONV, optimized by HOMER, for supplying a health clinic's demand. Besides, excess electricity management techniques were compared, including FC/ELEZ/HT. The best design was PV (2.52 kW), DG (2 kW), BAT (2 kWh), and CONV (1.66 kW), which resulted in COE and RF of 0.105 \$/kWh, and 30 %, respectively. Besides, involving the FC/ELEZ system could reduce excess electricity from 32.5 % to 6 %, accounting for fuel limitation. The estimated breakeven distance ranged from 1.2 km to 5.8 km, which means this off-grid MG can be developed and applicable for small demands. Ghenai and Bettayeb [20] sized and optimized a hybrid (PV/FC/DG/BAT) microgrid via HOMER software to meet a university building's demand with high RF and low COE and CO_2 emission. The results proved the ability of the proposed microgrid to fulfil the aim (RF = 66.1 %, COE = 0.092\$/kWh, and CO_2 = 24 ton/GWh) and the demand was met 73 % (PV), 24 % (FC), and 3 % (DG). Another study was presented by Jahangir MH et al. in [21] to electrify a residential community village in Iran (which is currently electrified from the Iranian national grid) using a standalone hybrid renewable energy system. Author employed PV, WTs, batteries, and biogas generator. The optimization results clarified that the (PV/WTs/BioGas generator/BAT) microgrid is the most cost-effective energy system with negligible CO_2 emissions. The authors also performed sensitivity analysis to check the performance of the optimal system configurations considering changes on biomass rate, biomass price, and inflation rate. The outcomes revealed that the COE is directly related with the inflation rate and biomass price while increasing the biomass price reduces the biogas generator production. However, the performance of the batteries was not analysed, and the change of the batteries price was not considered in the sensitivity analysis.

In addition, Roth et al. [22] developed a novel metamodel based on MILP with UML2 language as a competitive model with HOMER to assess the feasibility of PV/DG/BAT microgrid in standalone mode and for a tropical island's factory. The developed model showed enough flexibility to solve different problems from classical consumption (on-site) to self-use with partial injection. Shezan [23] optimized a WT/DG/BAT/CONV microgrid for a small Malaysian community (daily demand = 115 kWh, and peak = 11.2 kW) via iHOGA, DIgSILENT, and Matlab software. As reported, the COE and NPC were lowered to 0.05\$/kWh and 77,286\$, respectively. The hybrid micro-grid reduced COE, NPC, and CO_2 emissions by 73.5, 72, and 82 %, respectively, compared to another Malaysian traditional power plant. Das et al. [24] utilized the water cycle algorithm (WCA) algorithm to size off-grid microgrid (PV/BGG/PHES/BAT/CONV) for an Indian radio station. The results showed that WCA rapidly converged, had lower skewness and kurtosis, and attained better design solutions than moth-flame optimization (MFO) and GA. The optimum design, considering WCA, had NPC (\$0.8133 million) and COE (0.4864\$/kWh), and it consisted of a PV panel (69.2 kW), a BGG (16 kW), 21 batteries, a CONV (30 kW), and reservoir (2081.5 m^3). The NPC and COE values varied with load loss probability (LLP) values reaching \$0.693 million and 0.3685\$/kWh, respectively, at LLP of 0.05. Concerning irrigation electrification, Haffaf et al. [25] conducted a multi-criterion investigation using HOMER software to optimize a PV/DG/BAT/CONV microgrid for a water irrigation system with 40 kWh daily demand. Different scenarios were compared, among which the hybrid system was the most attractive and optimal one. Besides, the techno-economic comparison was conducted with and without load shifting (LS). The optimal system, without LS, components to fulfil the demand mentioned above were PV panels (10 kW), DG (7 kW), 23 BAT (4 kWh) and a CONV (9 kWh). The LS reduced the NPC, COE, and CO_2 emissions by 20, 19.25, and 10.7 %, respectively. In addition, 13 BAT and a DG were saved with an RF increase from 78.2 % to 80.4 %. Elkadeem et al. [26] proposed a new systematic framework for 3E (energy-economic-environmental) and sensitivity analyses of different five large-scale microgrids for electrification of agriculture and irrigation area in Sudan with a demand of 11,730 kWh/day/year. The microgrid design was conducted with the aid of the HOMER optimizer. Results revealed that hybridization of 2 MW solar array, 4.2 MW wind farm, 1.5 MW diesel genset 1, 0.5 MW diesel genset 2 (existing) 26.8 MWh battery packs and 2.9 MW converter is the cost-effective alternate at least NPC and COE of \$24.16 million and 0.387\$/kWh, respectively as well as the sizable share of renewables by RF of 95.12 % and thus 95 % reduction in CO_2 compared to diesel system.

In most recent studies on microgrid sizing, Tariq et al. [27] sized and optimized an autonomous PV/WT/DG/BAT microgrid for the water management of an isolated community of the indigenous Mayan region of Yucatan, Mexico, with different sizing optimization methods (single and multi-objective): HOMER, excel spreadsheet, and Non-dominating Sorting genetic algorithm II (NSGA-II). According to HOMER and spreadsheet, the optimal case is PV/DG/BAT (Li-ion) with COE (0.205–0.229\$/kWh) and NPC (\$70,000–\$79,000) having capacities of ∼ 17 kW (PV), ∼5kW (DG), and 44–48 kWh (BAT). Incorporating multi-objective optimization solved by NSGA-II, the COE and NPC declined by 0.86 % and 0.73 %, respectively, and RF reduced by 0.4 %. Ferahtia et al. [28] investigated a PV/FC/BAT building microgrid operated at two modes: standalone and grid-connected. Compared to other optimizers, the system was optimized via the bald eagle search algorithm (BESA). The suggested strategy obtained COE, high efficiency, and uplifted final charge state of 0.1577\$/kW, 87.395 %, and 33.268 %, respectively. Kiehbadroudinezhad et al. [29] utilized a division algorithm (DA) to formulate a multi-objective optimization of a standalone PV/WT/B/DG hybrid system to satisfy power-water cogeneration regarding low cost and impacts on human health and environment. A real case was considered (RO desalination plant in Larak Island, Iran). The investigation was conducted at various diesel fuel prices (DFP). For DFP = 0.2 and 0.5\$/L, the microgrid energy system based on PV/DG/BAT was the most cost-effective, whereas the PV/WT/DG/BAT system was the most cost-effective when DFP was 1\$/L. The DA was more accurate and faster than other algorithms (GA and artificial bee swarm). Table 1 summarizes the recent studies on the feasibility and optimal design of autonomous hybrid renewable microgrid systems.

Research gaps and study contributions

From the aforementioned literature survey, the main observations and research gaps can be summarised as follows:

a. Existing research recognizes the critical role played by local climate condition, renewable resources availability, and electricity consumption pattern in the selection of the proper mix of microgrid technologies.
b. It is well established from a variety of microgrid studies that a hybrid solar/wind/diesel system is the cost effective and reliable energy source compared to other fossil-fuel-based generation such as diesel in several locations.
c. Most existing body of research on microgrid have focused on the economic and/or technical feasibility for electrifying rural villages

Table 1
Recent studies on optimal design feasibility of standalone hybrid renewable microgrids.

Ref.	Site	Microgrid structure	Load sector	Solution method	Optimization criteria	Year	COE ($/kWh)
[13]	Peru	PV/DG/WT/BAT/CONV	Rural village	HOMER	NPC;COE	2020	0.4780
[30]	Benin	PV/DG/BAT/CONV	Rural village	HOMER	COE;CO2	2019	0.2070
[31]	Kenya	PV/WT/DG/BAT/CONV	Rural village	HOMER-MCDM	COE; CO2	2021	0.2800
[32]	Rwanda	PV/HKT/WT/CONV/BAT	Rural village	HOMER	NPC,COE;CO2	2020	0.0560
[14]	Nigeria	PV/WT/DG/BAT	Rural village	HOMER	COE,NPC;CO2	2019	0.1100
[33]	Sub-Saharan	PV/WT/DG/BAT/CONV	Rural village	MILP	COE	2019	0.6000
[34]	Morocco	PV/DG/BAT/CONV	Rural village	HOMER	COE;CO2	2020	0.2000
[35]	Pakistan	PV/DG/BAT/HYD/BIM/CONV	Rural township	HOMER-MCDM	NPC;CO2;JOB	2021	0.0740
[16]	Taiwan	PV/DG/BAT/CONV	Island area	HOMER	NPC, RF;COE	2020	0.3569
[36]	Egypt	PV/WT/DG/PHS/BAT/CONV	Desalination plant	HOMER-FMCDM	NPC;COE,RF;CSF	2021	0.1010
[17]	Germany	PV/WT/DG/BG/BAT/CONV	Island	HOMER	NPC;COE	2020	0.2960
[37]	Pakistan	WT/DG/BAT/CONV	Household	HOMER	NPC;COE	2019	0.3090
[38]	Nigeria	PV/WT/DG/BAT/CONV	Household	HOMER	COE;CO2	2019	0.5105
[39]	Algeria	PV/DG/BAT/CONV	Household	PSO	NPC;COE	2020	0.2000
[18]	Hong Kong	PV/WT/BAT	High-rise building	TRNSYS	COE	2020	0.2230
[19]	KSA	PV/FC/CONV/ELZ/H2T	Healthcare clinic	HOMER	COE	2021	0.1050
[20]	UAE	PV/FC/DG/BAT/CONV	University building	HOMER	CO2;COE	2019	0.0920
[26]	Sudan	PV/WT/DG/BAT/CONV	Agriculture facilities	HOMER	NPC;COE	2019	0.3870
[25]	Algeria	PV/DG/BAT/CONV	Water irrigation	HOMER	NPC;COE	2021	0.2960
[40]	Egypt	PV/WT/DG/BAT/CONV	Desalination plant	HOMER	NPC;COE	2020	0.1070
[41]	Egypt	PV/WT/DG/BAT/CONV	Urban community	MATLAB	NPC;COE;CO2	2020	0.2262
[42]	Egypt	PV/WT/DG/BAT/CONV	Urban city	HOMER	NPC;COE	2020	0.1500
[43]	China	PV/DG/BAT/CONV	Housing estate	HOMER	NPC;COE	2019	0.4800
[44]	KSA	PV/WT/DG/BAT/CONV	Residential community	HOMER	COE;NPC	2021	0.3460
[23]	Malaysia	WT/DG/BAT/CONV	Domestic community	iHOGA	COE;CO2	2021	0.0500
[24]	India	PV/BG/PHS/BAT/CONV	Radio station	MFO-WCA	NPC;COE	2019	0.4073
[27]	Mexico	PV/WT/DG/BAT/CONV	Wastewater treatment plant	HOMER-NSGA-II	NPC;COE;CO2;RF	2021	0.2030
[28]	Algeria	PV/FC/BAT/CONV	Commercial building	BESA	OPEC	2022	0.1570
Current	**Egypt**	**PV/WT/BAT/CONV**	**Water irrigation**	**HOMER**	**NPC;COE;RF**	**2022**	0.1190
Average COE							0.2502

Abbreviations: **BESA** (Blade eagle search algorithm); **OPEC** (Operating cost); **DA** (Division algorithm); **SCOW** (Specific cost of the water).

[13,14–30–35], households [37–39], remote areas [16,17,24] or urban communities [18–23–41,42–44].

d. So far, there has been no detailed investigation of the role of microgrids in the context of agriculture and irrigation sector, except [25] in Algeria and [26] in Sudan.
e. To the author's knowledge, a systematic understanding of how fully renewable microgrid contributes the sustainable energy development for the agriculture sector in Egypt.

This study covers the existing literature gaps through the following main contributions:

1. To perform a techno-economic simulation and sizing optimization of a proposed autonomous microgrid with 100 % hybrid renewable (PV/WT/BAT/CONV) for a real case study of agricultural and irrigation area in Egypt with the aid of the HOMER Pro optimizer.
2. To evaluate and analyze the feasibility, and performance of the proposed microgrid system in terms of energy and cost dimensions in comparison with four power generation alternatives: standalone PV/CONV (the existing real system), hybrid PV/BAT/CONV, WT/BAT/CONV, and standalone DG.
3. To conduct a parametric analysis aimed at examining the response of the optimal microgrid design against the uncertainty of seven key input parameters, including the capital cost of PV units, WTs and batteries, project lifetime, discount rate, solar irradiance, and wind speed.
4. To analyze the impact of different values for the maximum annual capacity shortage on the microgrid financial indicators and optimal microgrid configuration via reliability analysis.
5. To emphasizes the importance of solar and wind energies in the sustainable growth of agriculture and in achieving Egypt's vision 2030. Considering that Egypt possesses sunny weather and high wind speeds, exploitation of such local resources is essential for a national sustainable energy mix through efficient planning of green generation and gradually averting the reliance on fossil fuels.

Paper organization

In addition to the introduction presented in Section 1, the materials and proposed methods are introduced in Section 2, including the detailed modelling of all system's components. Besides, Section 3 introduces the description of the investigated case-study while the obtained simulation, optimization, and parametric analysis results are presented in Section 4. Finally, Section 5 summarizes the most important conclusions from the study.

Materials and method

This study investigates the feasibility and analysis of hybrid renewable-based microgrids' performance to meet the irrigation system's electricity demand. The techno-economic simulation and optimization are performed using HOMER Pro optimizer for five microgrid configurations, namely (1) PV/CONV (the existing system) as the base-case, (2) hybrid PV/BAT/CONV, (3) hybrid WT/BAT/CONV, (4) hybrid PV/WT/BAT/CONV, and (5) standalone DG for fair comparison and assessing the carbon saving benefits over the conventional supply option. The schematic representation of each microgrid is shown in Fig. 1. This section presents the microgrid components modelling, optimization method, design criteria, and constraints.

Modeling of the proposed microgrid

In this section, the mathematical modelling of various elements for the proposed microgrid system is presented in addition to the economic and technical specifications.

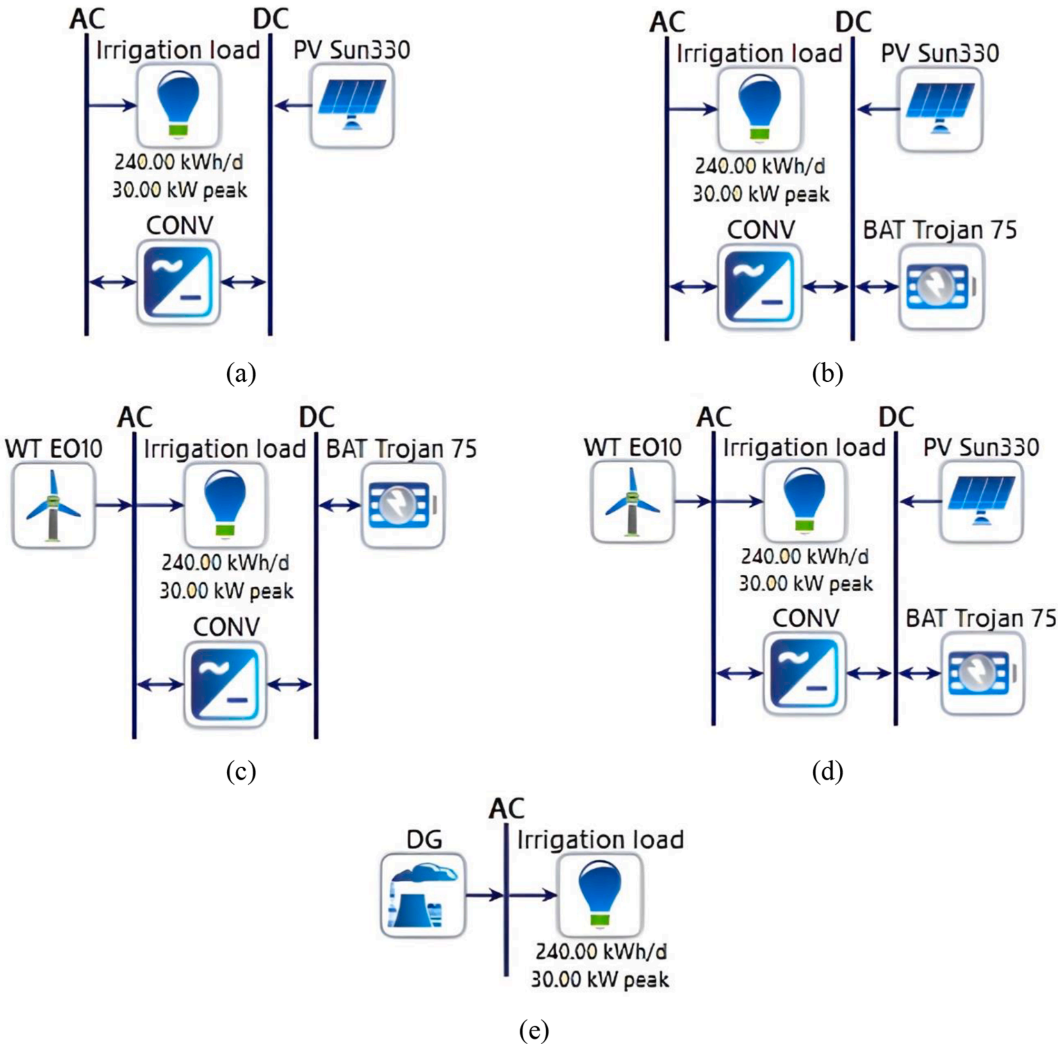

Fig. 1. Schematic representation of the investigated microgrid scenarios (a) existing PV/CONV, (b) PV/BAT/CONV, (c) WT/BAT/CONV, (d) PV/WT/BAT/CONV, and (e) standalone DG.

Solar photovoltaic

Solar PV modules are one of the most widespread technologies used for producing power from sunlight. The output electricity of solar PV system mainly depends on the value of solar radiation and ambient temperature as follows [45]:

$$P_{PV}(t) = P_{PV,r} \cdot f_{PV} \cdot \left(G_T(t)/G_{T,STC}\right) \cdot \left[1 + \beta_r \cdot \left(T_{PV}(t) - T_{PV,STC}\right)\right] \tag{1}$$

$$T_{PV}(t) = T_{amb}(t) + \left[(T_{NOCT} - 20)/G_{T,STC}\right] \cdot G_T(t) \tag{2}$$

The utilized PV model's technical and cost data are given in Tables A1 and A2 in the Appendix.

Wind turbine

The wind turbine technology is utilized to convert dynamic-wind energy into electricity. The power generated from the WT essentially depends on the wind speed profile at the installation site and turbine‘s specifications according to Eq. (3) [46]. As the elevation of the turbine rotor from the ground (i.e., hub height of the WT which is 20 m for the selected WT model) is not the same as the anemometer height, the actual wind speed is adjusted from the measured wind speed at anemometer height based on Eq. (4) [47]. It is worth to clarify that the anemometer represents an instrument that installed for measuring the speed of the wind at a specific site. In our study, the anemometers are installed specifically to determine wind power potential are placed at height equals 50 m. The wind actual wind speed Considering the site's air density, the actual output power can be calculated by Eq. (5) [48]:

$$P_{wt\_STP}(t) = \begin{cases} 0 & if\, V_{w,h}(t) \le V_{in} \\ P_{wt,r} \cdot \left(\left({V_{w,h}}^3(t) - V_{in}^3\right)/\left(V_r^3 - V_{in}^3\right)\right) & if\, V_{in} \le V_{w,h}(t) \le V_r \\ P_{wt,r} & if\, V_r \le V_{w,h}(t) \le V_{off} \\ 0 & if\, V_{w,h}(t) \ge V_{off} \end{cases} \tag{3}$$

$$V_{w,h}(t) = V_{anem}(t)\cdot\left[\left(\ln\left(h_{hub}/h_{ref}\right)\right)/\left(\ln\left(h_{anem}/h_{ref}\right)\right)\right] \quad (4)$$

$$P_{wt,G}(t) = P_{wt,STP}(t).\left(\rho_{act}/\rho_{air,ref}\right) \quad (5)$$

The specifications of the WT adopted in this study are given in Tables A1 and A2 in the Appendix.

Diesel generator

The crucial factor in launching any standalone system is reliability. In this study, diesel genset is utilized to reimburse any energy shortage due to the intermittency nature of solar and wind energies. The model type and price of the used diesel is provided in Tables A1 and A2 in the Appendix. Thankfully, diesel can offer improved flexibility without the cost or complexity of energy storage [49]. The diesel fuel consumption (F_c) was computed according to Eq. (6), while the electrical efficiency of the diesel genset (η_{gn}) can be expressed as in Eq. (7) [25].

$$F_c(t) = f_{gr}\cdot P_r + f_{go}\cdot P_g(t) \quad (6)$$

$$\eta_{gn} = 3.6\cdot P_g / \left(\rho_f\cdot F_c\cdot H_{lhf}\right) \quad (7)$$

Battery storage

Battery packs can sustain the load demand for more prolonged intervals of reduced solar or wind availability, such as night times and unwind intervals. This is to maintain the load demand and maximize the utilization of RE resources. The battery rated capacity CBAT (kWh) capacity can be calculated using Eq. (8). Also, the amount of stored energy in the battery at time t is obtained as expressed by Eq. (9) in case of battery charging and Eq. (10) in case of battery discharging [50,19].

$$E_{BAT,r} = E_L\cdot\gamma_{DA}/(\eta_{CONV}\cdot\eta_{BAT}\cdot DOD_{max}) \quad (8)$$

$$E_{BAT}(t) = (1-\alpha)\cdot E_{BAT}(t-1) + \left[\left(P_g(t) - P_L(t)/\eta_{CONV}\right)\cdot\eta_{BAT}\right]\cdot\Delta t;\ \text{charge} \quad (9)$$

$$E_{BAT}(t) = (1-\alpha)\cdot E_{BAT}(t-1) - \left[\left(P_L(t)/\eta_{CONV} - P_g(t)\right)/\eta_{BAT}\right]\cdot\Delta t;\ \text{discharge} \quad (10)$$

The specification of the utilized battery type is shown in Tables A1 and A2 in the Appendix.

System converter

A power converter is essentially for AC/DC energy conversion and vice versa. The capacity of the converter is designed to be greater than evacuated power from AC or DC sides as follows [9]:

$$\eta_{DC/AC}\cdot[P_{PV}(t)] \le P_{CONV} \quad (11)$$

$$\eta_{AC/DC}\cdot[P_{diesel}(t) + P_{WT}(t)] \le P_{CONV} \quad (12)$$

Optimization method for microgrid feasibility and design

In this work, Hybrid Optimization of Multiple Energy Resources (HOMER) Pro software [51], was utilized to perform the techno-economic optimization of the proposed microgrid under real renewable potential and irrigation load data. Fig. 2 shows the flowchart of the design procedure using HOMER Pro. First, a pre-feasibility assessment of the local climate data, cost parameters, technical specifications, and electricity demand is conducted. Then, identify the investigated microgrid configurations and study their techno-economic feasibility-one by one through one-minute energy balance calculation over a year. In each optimization loop, the optimal value of the decision variables (BAT capacity, WT number, PV array size, CONV rating or DG size) is optimized by minimizing the NPC. When the precision convergence of the NPC is achieved and all feasible configurations are analyzed, the algorithm generated a list of feasible solution ranked based on their economic performance. It is worth mentioning that the energy dispatch algorithm while optimizing the microgrid design is based on a load-following strategy. This strategy maximizes renewables instead of diesel (if any). The main priority of the system's DG is to fulfill the primary load only at the time when enewable energy fails. It is left to the PV and WT to serve other surplus power for serving other objectives such as battery charging. The reader can refer to Ramesh and Saini [53] for more details about the load following (LF) strategy.

Design optimization criteria

- **Net present cost (NPC, \$):** It is the present value of the stream of costs the system incurs over its lifetime, minus the present value of all the revenue it earns over its lifetime. It can be calculated by summing the total discounted cash flows in each year of the project lifetime [31]. In this paper, the NPC is used as the primary objective function to be optimized (minimized) and formulated as given below [31]:

$$\begin{aligned}\text{Obj Fn} &= \min(\text{NPC}) \\ &= \min\left\{\text{CAPC} + \left[\left(\sum_{n=1}^{N}\text{OPEC}_n + \text{FLC}_n + \text{RPC}_n - \text{SVC}_n\right)\Big/(1+d)^n\right]\right\}\end{aligned} \quad (13)$$

$$d_r = \left(d_r' - f_r\right)/(1+f_r) \quad (14)$$

where CAPC is the total capital of the microgrid system (\$), $OPEC_n$, FLC_n, RPC_n, SVC_n are operation, fuel, replacement, and salvage costs (\$) of the microgrid system in a year n, while d_r is the annual real discount rate (%). Also, N is the project lifetime (year), d_r' is the nominal interest rate to borrow money (%), and f_r is the expected inflation rate (%).

- **Cost of energy (COE, \$/kWh):** It can be regarded as the minimum cost at which electricity produced by the system must be sold in order to achieve breakeven over the project lifetime [54]. It is expressed as the average cost per kWh of useful electrical energy generated by the microgrid system as follows [55].

$$COE = C_{Tot,\,an}/E_g \quad (15)$$

$$C_{Tot,\,an} = NPC \bullet \left[d_r \bullet (1+d_r)^N\right]/\left[(1+d_r)^N - 1\right] \quad (16)$$

where E_g is the annual useful electrical energy generated by the microgrid (kWh/yr) and $C_{Tot\ an}$ is the total annualized cost (\$/year).

- **Capacity shortage fraction (CSF)** refers to the probability of an energy deficit occurrence in the system during a certain period. The CSF is mathematically represented as the total capacity shortage divided by the total electrical served load during a year as given in Eq. (17) [35], where E_L is the total annual load served (kWh/year). This criterion is applied as a reliability measure to investigate and assess service continuity. The value of ZERO (i.e., CSF% = 0) refers that the load be fully fulfilled, means that the annual useful electrical energy generated (E_g) equals the total annual load (E_L). In contrast, the value of ONE and (i.e., CSF% = 100) denotes, respectively the load will never be served, in which Eg will equal zero.

$$CSF = \left(E_L - E_g\right)/E_L \quad (17)$$

- **Total carbon emission (TCE, kg/year):** It is an essential measure to analyze the emission performance of the microgrid system in terms of the total carbon dioxide emitted by the diesel generator per year, and it is calculated as follows [56]:

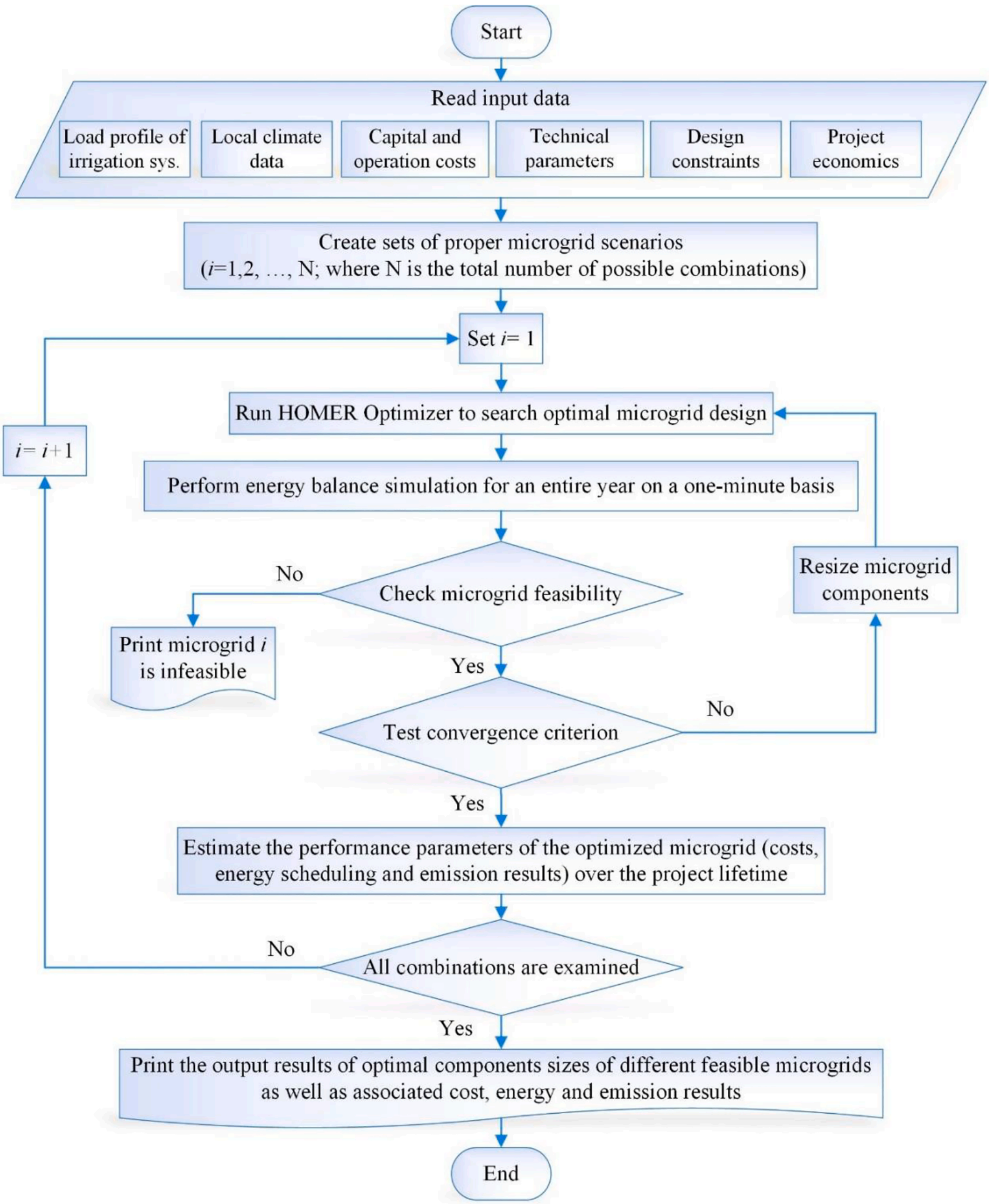

Fig. 2. Flow chart applied for the feasibility and optimization of fully renewable microgrid.

$$TCE = \sum_{t=1}^{8760} SE_{CO2} \bullet FC(t) \tag{18}$$

FC is the hourly fuel consumption (litre/hr), SE_{CO2} is the per litre diesel's specific emission of CO_2, and its value is 2.7 kg/litre.

Design optimization constraints

- **Load balance constraint:** The net power generation shall be equal to total load demand at any time. For example, Eq. (19) should be met in the case of a hybrid PV/WT microgrid.

$$P_{PV}(t) + P_{WT}(t) = P_L(t) \tag{19}$$

- **Battery constraint:** The State of Charge (SOC) of batteries and the power availability of battery charge/discharge must be within the safe boundaries at any time according to Eqs. (20) and (21) [57,58]:

$$SOC_{BAT,min} \leq SOC_{BAT}(t) \leq SOC_{BAT,max} \tag{20}$$

$$E_{BAT,r} \cdot (1 - DOD) \leq E_{BAT}(t) \leq E_{BAT,r} \tag{21}$$

Fig. 3. Fixed tilt angle ground-mounted PV Plant (photo for the existing real system).

- **Generation capacity constraint**: In the case of standalone DG microgrid, the output power of diesel as a dispatchable unit must satisfy its upper and lower limits [59]:

$$P_{DG,\,min} \le P_{DG}(t) \le P_{DG,\,max} \tag{22}$$

Case study

The deep well water pumping plant used to irrigate a grape farm on 30 acres in Sadat, Egypt (30.364365 N, 30.506903 E). The water level in the deep well is 55-meter below the sea level whereas the more well depth requires more power. The plant load consists of one submersible water pump of 30 hp rated power, 2900 r.p.m. rated speed, 100 m^3 hr^{-1} water flow at rated speed. The pump has seven stages of operation and is driven by a 30 hp induction motor. Under real conditions, the plant works around 8 h of daylight from 7:00 a.m. to 3:00p.m. and is supplied by a battery-less standalone solar PV microgrid. It is worth to mention that the hourly time pattern was obtained from the site survey. The monthly level of the load demand was constant according to the real operating condition of the pump plant. The solar system consists of six parallel strings; each string consists of sixteen series PV panels. Each PV panel is 330 Wp Suntech polycrystalline type. The PV array is ground-mounted with a 30-degree fixed tilt angle, as shown in Fig. 3. The average volume of water is 600 and 800 m^3/day for the winter and summer seasons, respectively.

A 30 hp solar inverter is used to convert the DC power from the solar plant into three-phase AC electricity. The inverter parameters were adjusted for the voltage/frequency control algorithm, which provides speed control at constant torque. This technique ensures that the inverter's output voltage is proportional to the frequency, maintaining a constant motor flux. When the input power from the PV array changes according to the unpredictable variation of solar irradiation, the voltage-dependent capacitance circuit of the inverter cause changes in the oscillation frequency, resulting in driving the induction motor with variable frequency. This allows the water pump to operate at variable speeds. This system ensures a continuous operation of the water pump directly on sun radiation with variable speed. Due to the weight of the water column located on the submersible pump, the inverter parameters are adjusted to achieve both the pump's soft start and soft stop. The pump is protected from working at over speed at maximum solar irradiation by adjusting the maximum frequency of the inverter output, and the pump is protected from working at under-speed at sunrise and sunset by adjusting the minimum frequency of the inverter output. The rated DC input voltage to the inverter is 600 V output AC voltage is 380 V. An enclosure is used to protect the inverter from fine dust and dewy since its IP is 21. The inverter and its enclosure panel are given in Fig. 4-a. It is worth mentioning that a combiner box with two fuses per string and a DC switch is implemented for protection purposes (see Fig. 4-b). The DC

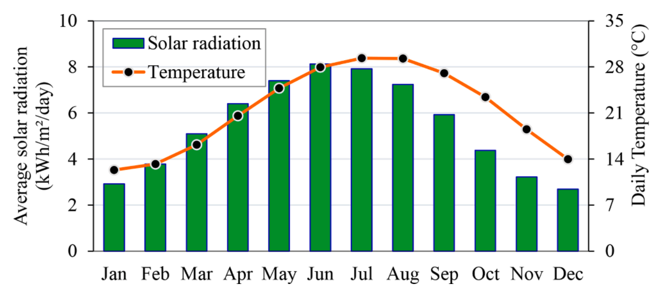

Fig. 5. Monthly average solar irradiation and daily temperature.

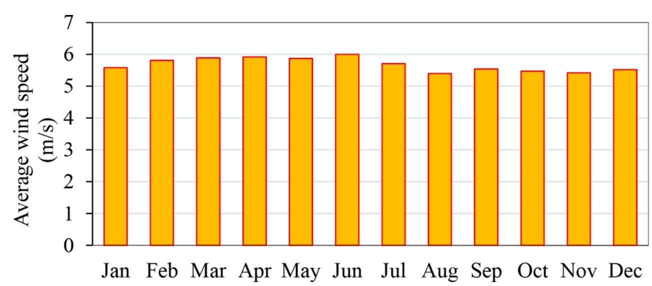

Fig. 6. Monthly average wind speed at 50 m height.

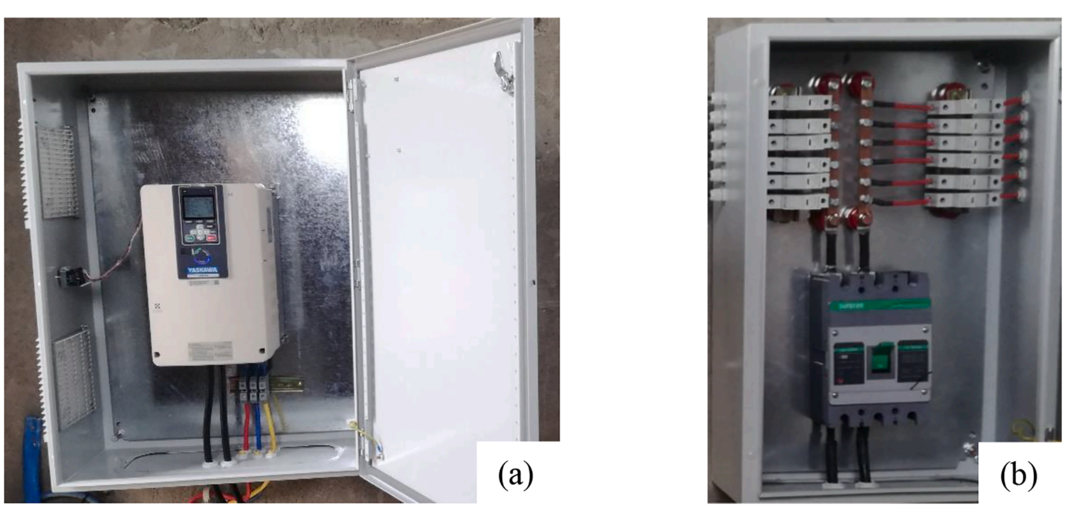

Fig. 4. (a) The inverter with its enclosure panel and (b) combiner box (photo for the existing real system).

Table 2
Optimization results of the investigated microgrid configurations.

Case	Optimal design capacities					Evaluation Criteria									
						Economic				Technical				Emission	
	PV kW	WT Qty	DG kW	BAT Qty	CONV kW	NPC $	COE $/kWh	OPEX $	CAPEX $	RF %	CSF %	SE %	UL %	Fuel L/y	CO_2 kg/y
1	32	–	–	–	30	27,849	0.0351	486	18,800	100	51.45	15.36	51.45	–	–
2	199	–	–	350	31	286,902	0.1761	5,179	190,521	100	0.099	71.68	0.099	–	–
3	–	15	–	500	34	507,952	0.3120	10,005	321,750	100	0.099	87.32	0.099	–	–
4	85	2	–	200	30	194,017	0.1190	3,868	122,032	100	0.098	61.69	0.098	–	–
5	–	–	30	–	–	36,359,4	0.2200	1,7925	30,000	0	0	0	0	26,806	70,173

fuse rate is 20 A, and it is used to protect the DC side of the inverter from short circuit currents. The DC switch is used to disconnect the DC power from the PV array during inverter maintenance time.

For assessing the renewable energy resources, the solar irradiation and wind speed profiles at the investigated site are given in Fig. 5 and Fig. 6, respectively, based on historical meteorological data sets available by NASA [60]. It can be observed from Fig. 5 that the solar radiation increases from January till reaching the maximum value in June then it is starting to decrease again every month to reach the lowest value in December. The months of May, June, and July have the highest radiations in the year. Meanwhile, the selected site has an approximate profile of constant wind speed as illustrated in Fig. 6. Fortunately, June has also the highest wind speed and April is the second month which has the second level of the highest wind speed. Therefore, in June it is expected to get high power from both solar system and wind generating system. Finally, the data illustrate that the study locality is advantageous and possesses an abundance of sunny weather with few cloudy days at annual average radiation of 5.43 kWh/m^2/day and high wind speeds at an annual average of 5.68 m/s. This makes an attractive location for solar and wind deployment.

Results and discussion

This section provides the results of optimal components' capacities and the economic, technical, and parametric analyses of the proposed microgrid. Based on the flow chart given above in Fig. 2, the techno-economic simulation and optimization are performed for various feasible microgrid configurations as listed below:

1. PV/CONV (the existing real system, as a base-case)
2. PV/BAT/CONV,
3. WT/BAT/CONV,
4. PV/WT/BAT/CONV
5. Standalone DG

The standalone DG configuration was considered for fair comparisons among the entire alternatives as most the existing systems are electrified from the main grid or from a standalone fossil fuel-based generator. The design optimization results are presented in Table 2 using a project lifetime of 25 years, a real interest rate of 8.25 % and an expected inflation rate of 5.70 % [61]. The table demonstrates the optimal sizes of every component coupled with each energy system scenario that satisfies the load demand with the different associated constraints. Besides, it reveals the different evaluation criteria, including the different economic, technical, and energy data. Both the NPC and the COE are considered the main driving factors to test the economic feasibility of each configuration. Renewable friction (RF), capacity shortage friction (CSF), surplus energy (SE), and unmet load (UL) are considered the criteria to evaluate the technical feasibility and energy performance of each energy system configuration. The fuel consumption and the resulting CO_2 emissions are also indicated for the energy scenarios that integrate diesel generators; however, in this study, the diesel generator was individually examined to represent the most commonly-used energy system for supplying such applications. Using HOMER Pro, hourly-based simulations were performed to examine the feasibility of the candidate energy systems. Each of the five scenarios is discussed and analyzed in the following subsections to demonstrate its technical, economic, and ecologic performance.

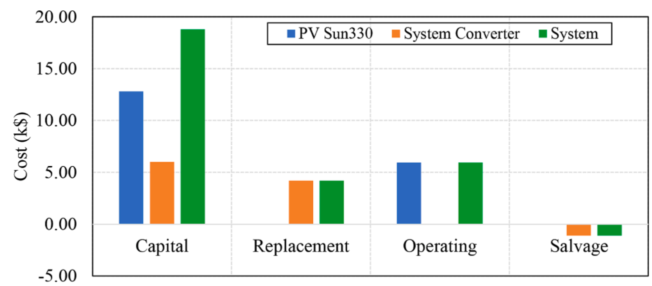

Fig. 7. Cost summary of the PV/CONV microgrid.

Standalone PV/CONV microgrid (existing system and base-case)

According to the design specification of the real field system, the PV and converter capacities are recorded at 32 kW and 30 kW, respectively. This system was firstly modelled and simulated in HOMER Pro to obtain in-depth performance statistics including the economic, the ecologic, and the technical aspects, and to highlight the pros and cons of the existing energy system for later comparison with the proposed and feasible alternatives. The techno-economic simulation results of this scenario revealed that the NPC is $27849, and the COE is 0.0351$/kWh. This system enjoys zero-emission due to the presence of PV only. This system's cost summary and cash flow are shown in Fig. 7 and Fig. 8, respectively. It can be recognized that the PV capital cost dominates the largest portion of the NPC in this system ($12,800), where the system converter requires nearly half PV's price ($6,000). Only the PV system requires an operating cost of $5,955. Besides, since the lifetime of both the PV system and the system converter are 25 and 15 years, respectively, the replacement costs go only for the system converter with $4,196.21, while the salvage costs will be about $1,102 at the end of the project lifetime. From an energy flow aspect, the PV was responsible for serving the total energy consumption of the pumping load (240 kWh/day and 30 kW peak) over a year, as described in Fig. 9. The PV system has a nominal capacity of 32 kW, which produces an annual production of 52,893 kWh/yr and mostly between 8:00 a.m. to 4:00 p.m. as illustrated in this system's annual production heat map in Fig. 10.

Fig. 11 shows the electrical energy dispatch for three consecutive days in April and an entire day in November to investigate the load balance at each step. From the figure, it can be clarified the contribution of the PV system in supplying the load demand during the specific periods. The figure also highlights that the existing system is not capable to satisfy the primary load demand as indicated from the unmet load and the total electrical load served curves. The simulation results reveal the high amount of unmet load (45,073 kWh/yr) that accounts for more than half of the electricity consumption need of the irrigation facilities.

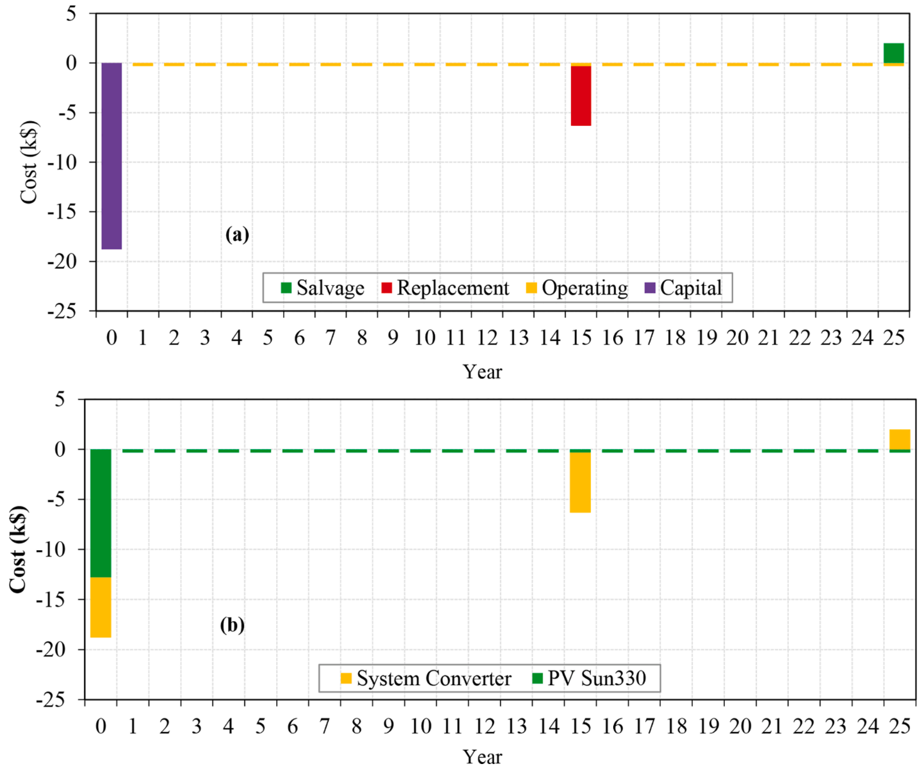

Fig. 8. Cash flow of the PV/CONV microgrid by (a) cost type (b) component type.

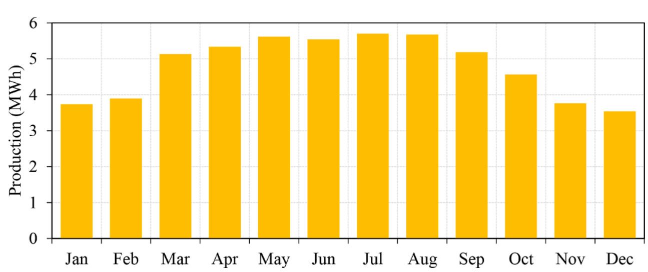

Fig. 9. Annual energy production summary of the PV/CONV microgrid.

This is attributed to the large CSF (51.5 %) of this configuration due to the PV system's single-use and under-sizing calculation. Furthermore, since this configuration does not contain batteries, a total amount of 8,128 kWh (15.4 % of the total electricity production) is wasted annually. In short, however, the best economic results of the PV/inverter system with the least NPC and COE among all configurations are unfeasible from technical aspects with the unreliable operation.

Hybrid PV/BAT/CONV microgrid (case 2)

In this case, a combination between the PV panels and the batteries is proposed to supply the load. The main contribution of the PV panels is during the daytime; meanwhile, the batteries banks provide the power to the load during the night and shad periods. Also, the battery system could store the surplus energy during the existence of a large amount of energy from the sun over the load demand. The system feasibility is tested using hourly-based simulations and optimization to get the optimal component sizes, achieving minimum NPC and COE while maintaining the load demand. The results of the optimized system are 199-kW for the PV panels, 350 batteries from Trojan SAGM type (7 strings in parallel with string size of 50 batteries), and a 30.6-kW converter. In this case, the NPC is $0.2869 million, and COE is 0.1761628 $/kWh. Fig. 12 shows the cost summary in terms of NPC for this system. It can be observed that the major portion of NPC is associated with batteries, where the capital cost represents $0.105 million, and the operating cost is $0.0651 million. The second portion of the NPC is ascribed to costs associated with PV panels which are $0.0794 million for the capital cost and $0.0369 million for operating cost. The rest of the NPC is for the system converter. The annual energy production from the PV panels and the batteries bank are given in Figs. 13 and 14, respectively. It is noticed that the PV is exploited and thus contribute

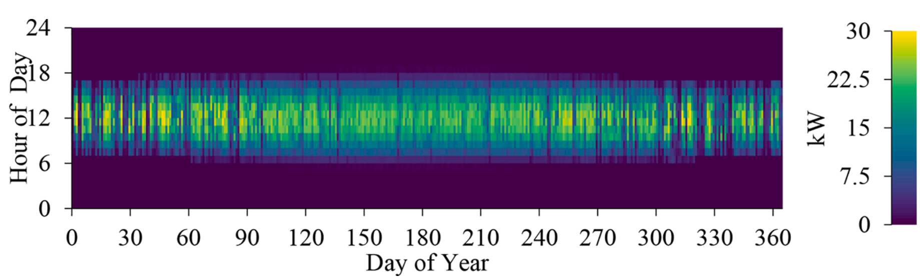

Fig. 10. Annual production heat map of the PV/CONV microgrid.

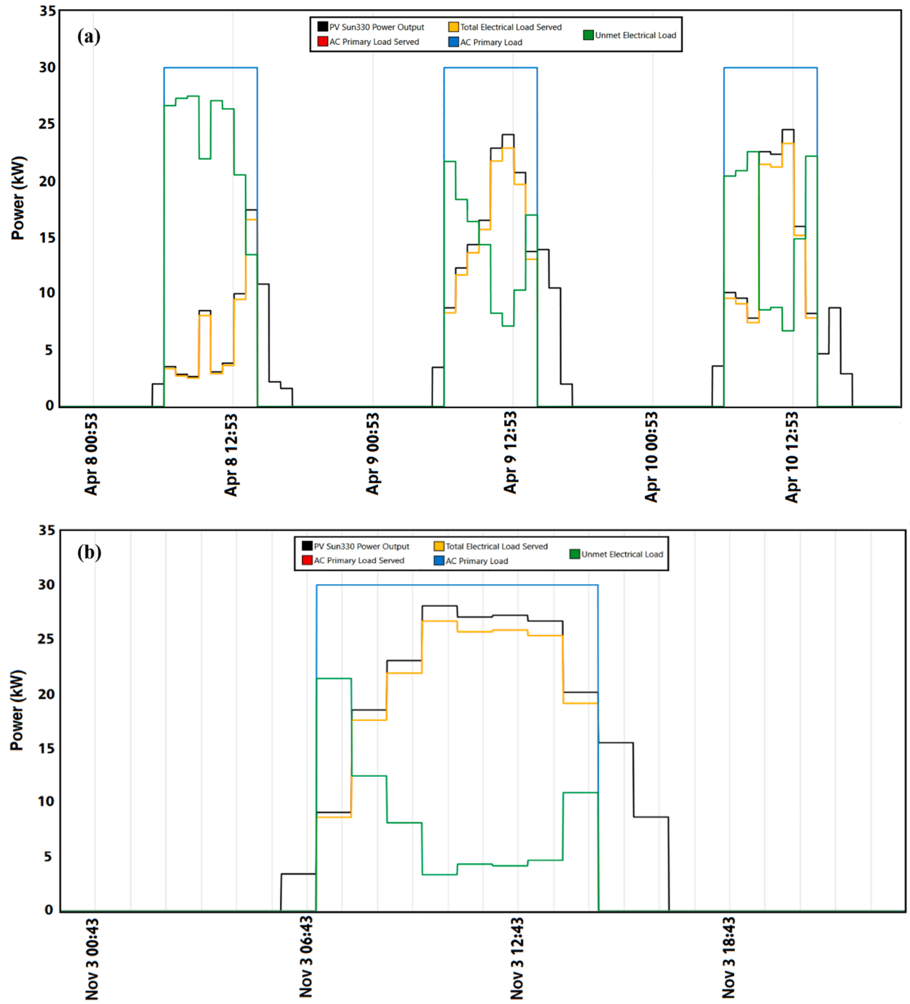

Fig. 11. Energy dispatch for the PV/CONV microgrid (a) three successive days (8, 9, and 10th April) (b) an entire day (3rd November).

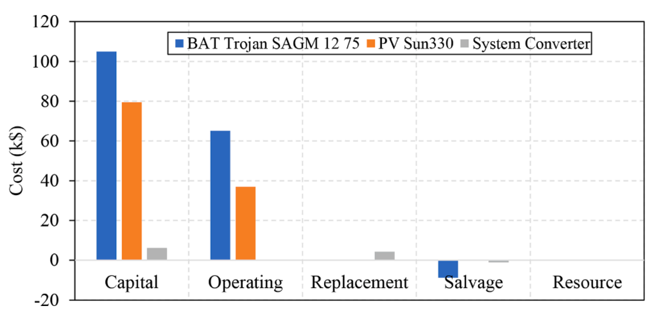

Fig. 12. Cost summary of the hybrid PV/BAT/CONV microgrid.

more power during the mid-day during peak sun hours, epically in the summer season. On the other side, the battery share or discharge occurs mostly in the winter months (January, November, and December) to continuously fulfil the load demand.

Hybrid WT/BAT/CONV microgrid (case 3)

A wind generator is employed with the battery storage system in this system. The batteries bank is used to indemnify the energy deficiency from the wind resource where the batteries supply the load with the required power in the event. On the other hand, the wind energy is large during high wind speeds, and the batteries bank becomes charging. After the techno-economic optimization, the system consists of 15 wind turbines from the Eocycle EO10 model, 500 batteries from Trojan SAGM 12

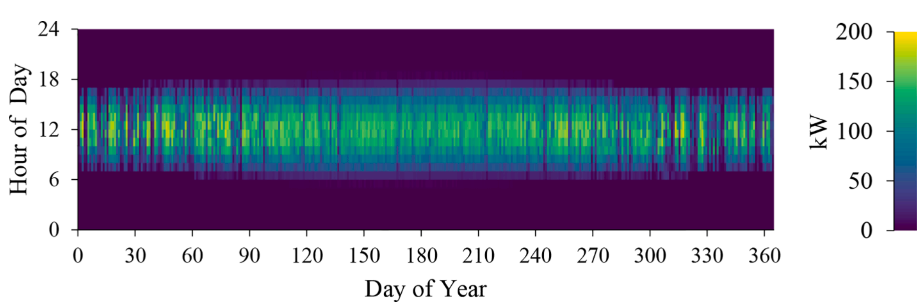

Fig. 13. Annual energy production summary of the PV units.

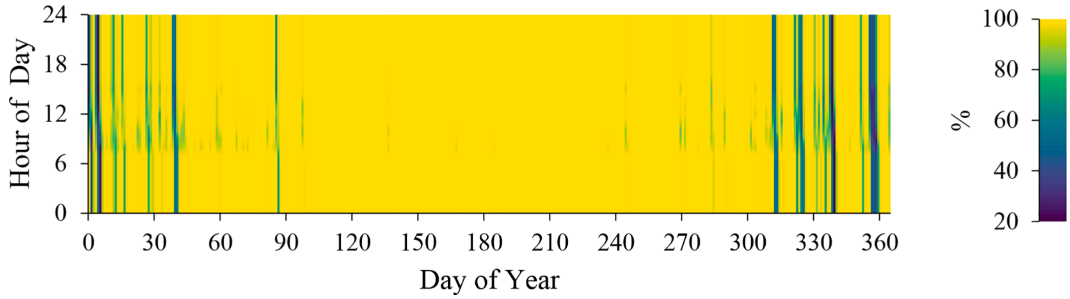

Fig. 14. Annual energy production summary of the batteries.

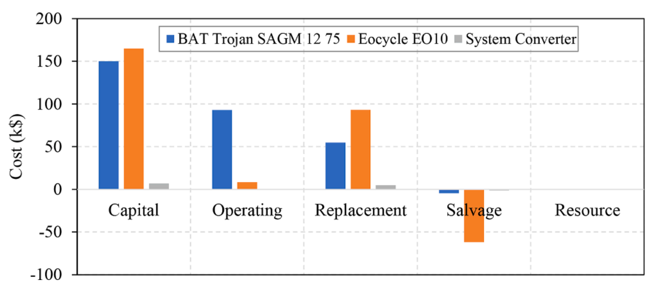

Fig. 15. Cost summary of hybrid WT/BAT/CONV microgrid.

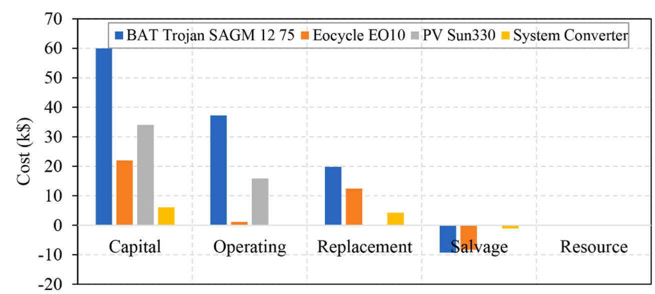

Fig. 18. Cost summary of hybrid PV/WT/BAT/CONV microgrid.

Table 3
NPC distribution in k$ of the WT/BAT/CONV system.

Element	Capital	Operating	Replacement	Salvage	Total
BAT Trojan SAGM 12 75	150.00	93.05	54.72	4.55	293.22
Eocycle EO10	165.00	8.38	93.12	61.99	204.50
System Converter	6.75	0.00	4.72	1.24	10.23
System	321.75	101.43	152.56	67.78	507.95

75 types (10 strings in parallel with string size of batteries) 33.8-kW power converter. Details cost for the optimal system in terms of NPC are demonstrated in Fig. 15. The NPC for this configuration is $0.5079 million and COE of 0.31189$/kWh. The determined values of component costs are given in Table 3. The annual energy production summary for the wind turbines and the batteries are shown in Figs. 16 and 17, respectively. It can be observed that the output power from the wind system varies during the year and every day in random based on the wind speed profile. Also, the battery is working at unwind periods to serve the unmet load and enhance the supply reliability of the overall system.

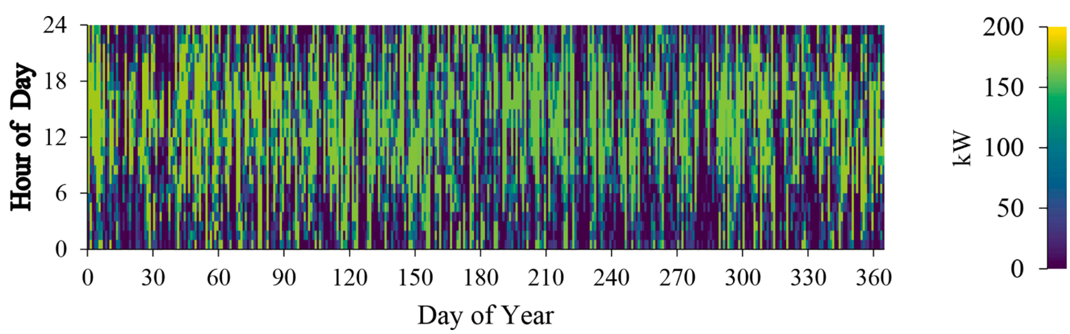

Fig. 16. Annual energy production summary of the WT.

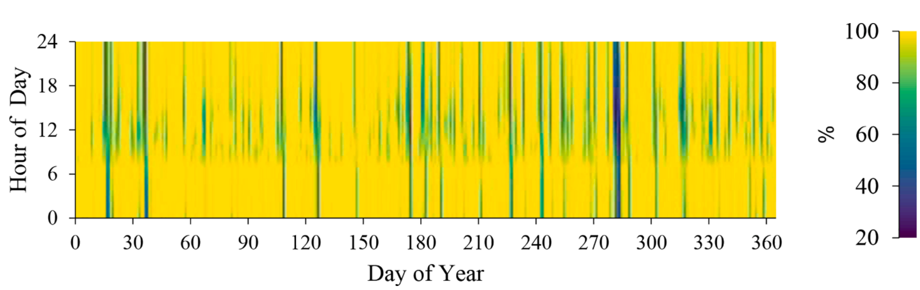

Fig. 17. Annual energy production summary of the batteries.

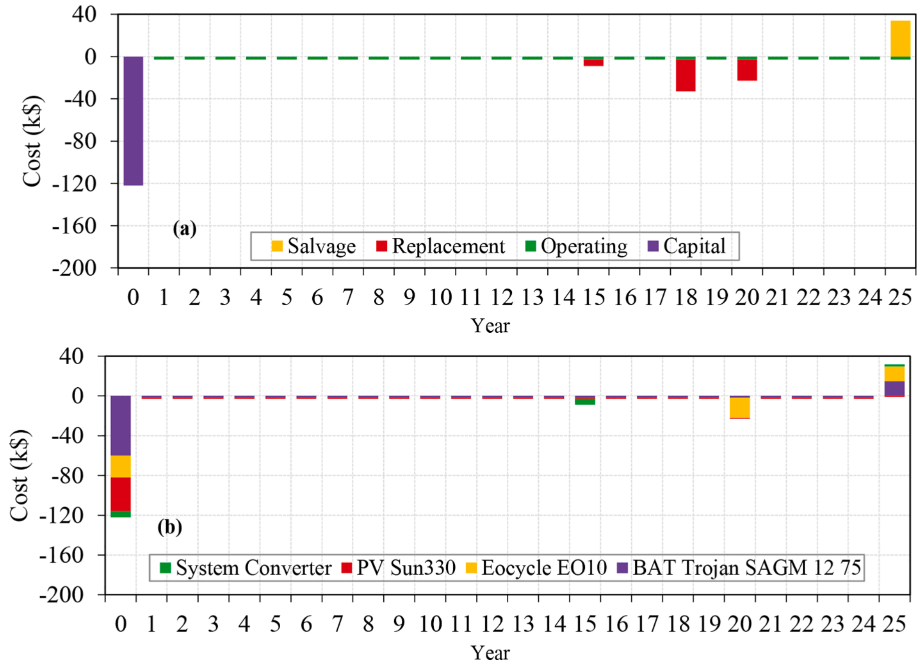

Fig. 19. Cash flow of hybrid PV/WT/BAT/CONV microgrid by (a) cost type (b) component type.

Table 4
NPC distribution in k$ of the PV/WT/BAT/CONV microgrid.

Element	Capital	Operating	Replacement	Salvage	Total
BAT Trojan SAGM 12 75	60.00	37.22	19.83	9.26	107.79
Eocycle EO10	22.00	1.12	12.42	8.27	27.27
PV Sun330	34.00	15.82	0.00	0.00	49.82
System Converter	6.03	0.00	4.22	1.11	9.14
System	122.03	54.16	36.46	18.64	194.02

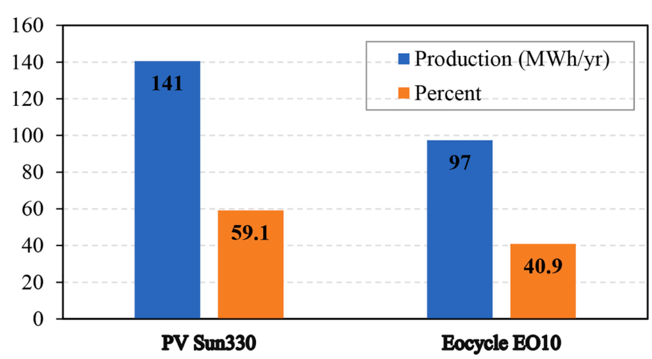

Fig. 20. Production summary of the PV/WT/BAT/CONV system.

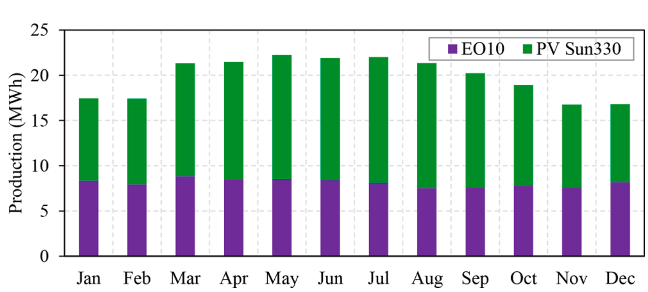

Fig. 21. The monthly average electrical output of the PV and the WT.

Hybrid PV/WT/BAT/CONV microgrid (case 4)

The proposed system, in this case, compromises multiple renewable energy sources (PV and wind turbine) to supply the investigated irrigation facilities. Besides, batteries bank to fulfil the energy storage requirement during the extra energy over the load requirements. The load is fed by the batteries bank in the shortfall periods of renewable energy resources. For this scenario, the optimal capacities are two Eocycle EO10 10-kW wind turbines, 85-kW PV panels, 50 batteries × 4 strings in parallel from Trojan SAGM type, and a 30-kW power converter. The summary and the cash flow for each part of this system are depicted in Fig. 18 and Fig. 19, respectively. Meanwhile, the NPC of this system is $0.194 million, and COE is 0.119$/kWh. It is seen from Fig. 18 and Table 4 that the major portion of the NPC is coming from batteries bank followed by PV panels Eocycle EO10 wind turbine. The production summary of the electrical power of this configuration is shown in Fig. 20. About 60 % of the total electrical production is contributed by the PV panels, while the remaining 40 % is from the wind turbine. This confirms the abundant availability of renewable resources of the studied location. The details on the average electrical output from the PV and the wind turbine at each month over a year is illustrated by Fig. 21.

In order to validate the load balance at each time step during the simulation, Fig. 22 shows the electrical energy dispatch for three consecutive days in August and an entire day in July. The figure demonstrates the contribution of each element in supplying the load demand during the specific period. Besides, it reveals the behaviour of the battery through the battery's SOC in which the power was transferred from the battery to the load (discharging mode) during the periods the production from both the PV and WT system became insufficient. The obtained results confirm the success of the proposed hybrid microgrid system to fulfil the load demand at each hour with a high level of service reliability.

Stand-alone DG (case 5)

In this case, a 30-kW diesel generator is used to supply the entire electrical load with a total annual production of 87,600 kWh, respectively. As seen before in Table 2, the NPC and COE of this system are recorded the highest values of $363,594 and 0.223$/kWh, respectively,

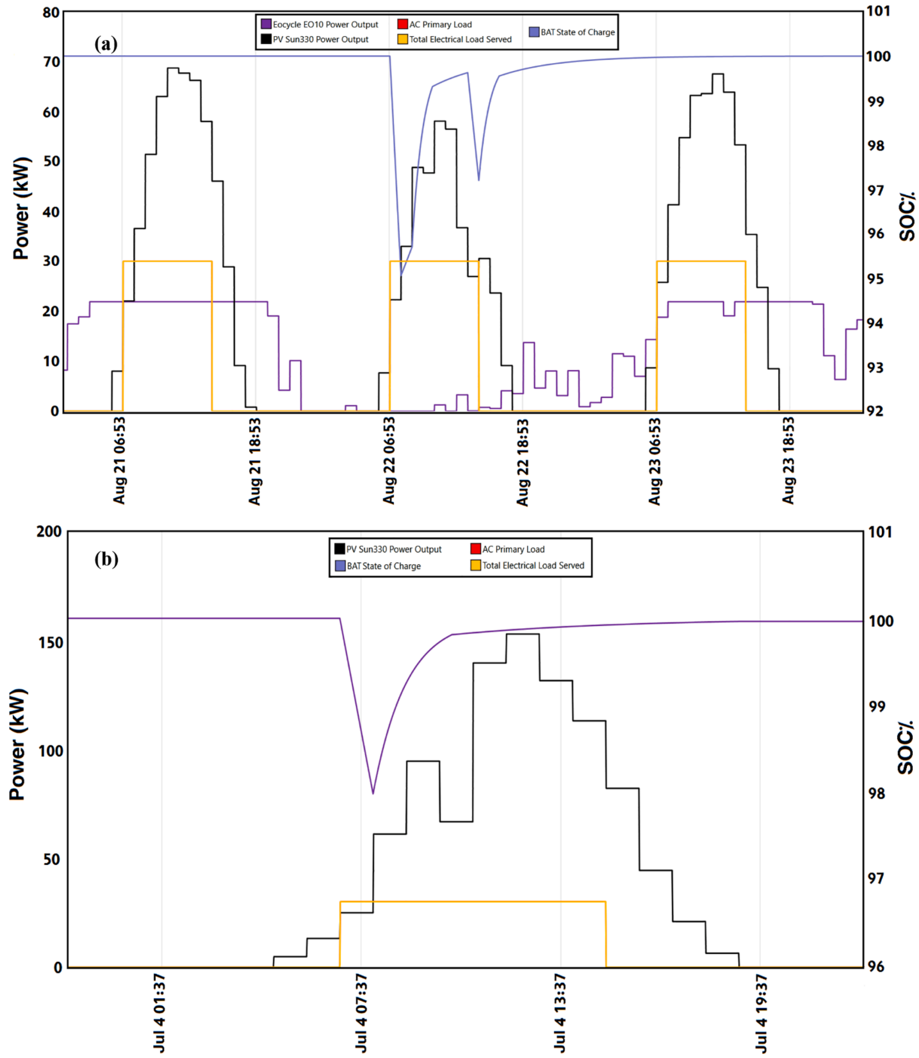

Fig. 22. Energy dispatch for the PV/WT/BAT/CONV microgrid in (a) three successive days (21st, 22nd, and 23rd August) (b) an entire day (4th July).

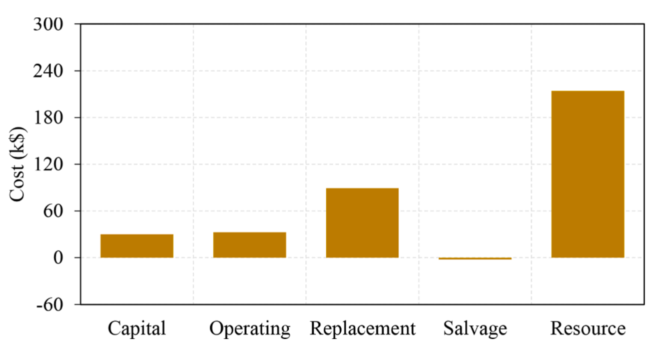

Fig. 23. Cost summary of the diesel-only system.

denoting the negative economic feature of this case. The cost summary for the NPC is illustrated in Fig. 23. The highest portion of NPC is caused by the diesel fuel price, which accounts for 58.8 % of the system NPC. Fig. 24 shows the monthly average electrical output using a diesel-only system. From the technical point of view, the system succeed to meet the entire load demand during 8760 hr with the minimum capacity shortage. However, it emits 70173 kg/year, making it an unfavorable option from the ecological aspect.

Comparison among various microgrid scenarios

This section compares the various five microgrid configurations based on their economic profile. In order to offer a readable and decipherable comparison, Table 5 and Fig. 25 show the results of different cost types of the studied microgrids and the final ranking based on the NPC value. It can be observed from Fig. 25 that the WT/BAT/CONV hybrid microgrid dominates the highest values for all cost items with $321,750, $101,425, $152,555, and $67,778 for capital operating, replacement, and salvage, respectively. With the exclusion of the non-technically feasible solution based on standalone PV/CONV microgrid due to its poor reliability level, it can be recognized that the PV/WT/BAT/CONV hybrid microgrid is the most cost-effective to meet the load demand of the irrigation facility under study. This system has the lowest capital, operating, and replacement costs of $122,032, $54,157, and $36,463, respectively. This is mainly attributed to the dramatic and continuous drop of PV modules price, which has fallen by 85 % from 2010 to 2020 [65] due to the economies of scale, research development, and learning-by-doing mechanisms [62]. On the other side, the diesel-only system performs with $30,000 capital cost and $246,614.34 operating cost due to its fuel consumption of 26,806 L/year. Furthermore, Finally, the COE is illustrated in Fig. 26, where it can be noticed that, by excluding the base-case, the WT/BAT/CONV system has the highest value among all configurations with 0.312$/kWh while the PV/WT/BAT/CONV system has the minimum COE with only 0.119$/kWh.

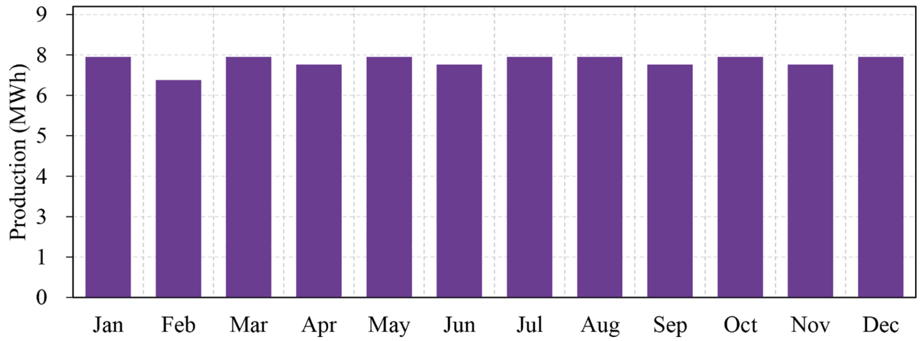

Fig. 24. Average electrical production of the diesel-only system.

Table 5
Cost summary in k$ and final rank of the microgrid configurations.

Metric/System	DG	PV/BAT/CONV	WT/BAT/CONV	PV/WT/BAT/CONV	PV/CONV
Capital	30.00	190.52	321.75	122.03	18.80
Operating	246.61	102.08	101.43	54.16	5.96
Replacement	89.18	4.28	152.56	36.46	4.20
Salvage	2.20	9.97	67.78	18.64	1.10
NPC	363.59	286.90	507.95	194.02	27.85
Rank	3	2	4	1	Reference
Notes	Not-eco-friendly	Not-economical	Non-economical	Most sustainable	Unreliable

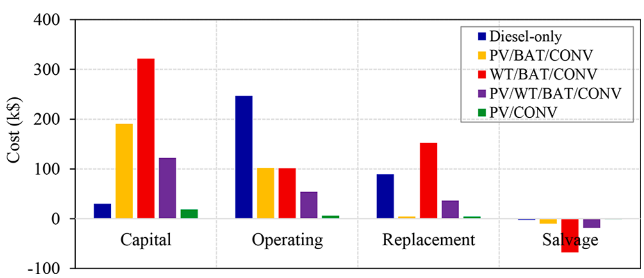

Fig. 25. Cost summary of the investigated microgrid configurations.

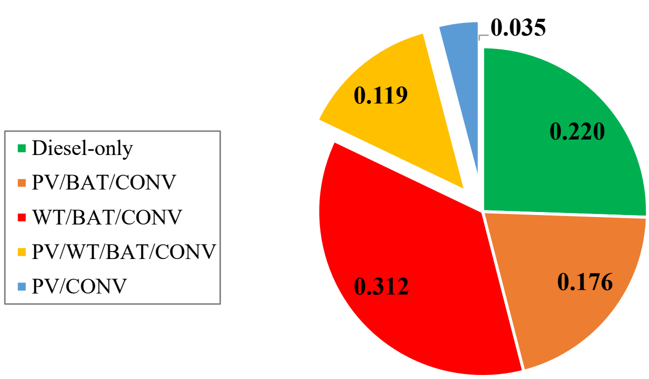

Fig. 26. Cost of energy in $/kWh for investigated microgrid configurations.

This confirms the superiority of the proposed microgrid to maintain the load demand at the lowest electricity price.

Parametric analysis for the microgrid system

Parametric analysis, also known as sensitivity analysis, analyses the impact of distinct statistical or physical parameters or even both of them on a problem's solution. Parametric analysis is crucial for exploring the energy system design feasibility and its possible/future behaviour against input parameter variations. Herein, the parametric analysis is performed to examine the response of the optimal energy system against the change of seven input parameters, as shown in Table 6.

Table 6
Parameters' change employed for the parametric analysis.

Input parameter	Ref. value	Change ratio
PV capital cost	400 $/kW	−50 %
WT capital cost	11,000 $/turbine	−50 %
Battery capital cost	300 $/battery	−50 %
Project lifetime	25 years	± 20 %
Discount rate	8.25 %	± 20 %
Solar irradiation	5.43 kWh/m^2/day	± 10 %
Wind speed	5.68 m/s	± 10 %

These seven inputs were delicately chosen based on the following motives:

- Since the rapid development and great technological advancement in material science impact both renewable energy and energy storage technologies, decreasing their costs. The effect of lessening the capital cost of the PV, WT, and battery storage systems by 50 % is assessed.
- Since the discount rate primarily varies with the economic condition of the country of the project, two prospects are performed once by raising the discount rate by 20 % and once again by lessening its value by the same ratio.
- Since the project lifetime is considered an important factor that influences the economic performance of all system's components, changing the duration of the project lifetime directly impacts the economic behaviour of the whole system. The project lifetime is varied by ±5 years of its reference period.
- Due to the uncertain nature of both solar irradiance and wind speed, which mainly depends on the project location, it directly affects the optimal capacities of both PV and WTs and impacts the project's economic performance. A change of ±10 % away from the annual average value in solar irradiance and wind speed is performed.

Fig. 27 demonstrates the response of the NPC and COE against the variations in the seven considered parametric variables. From Fig. 27(a),

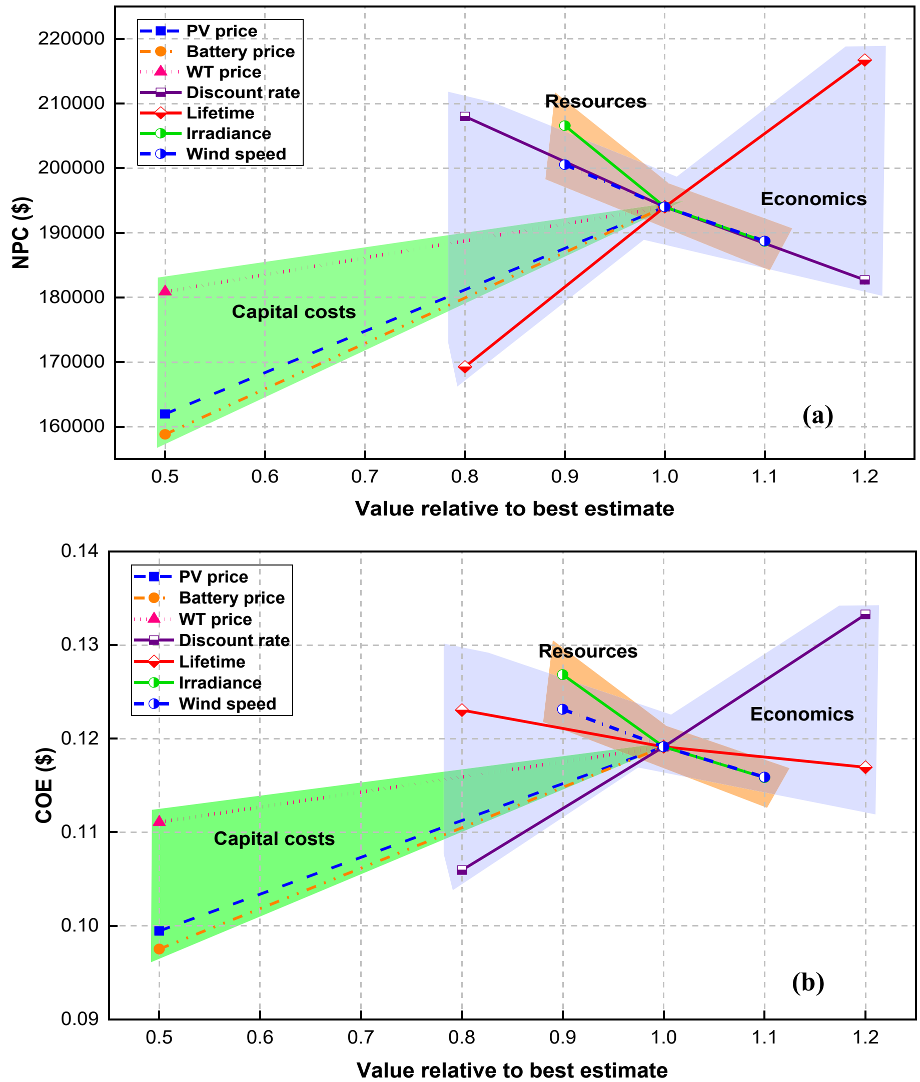

Fig. 27. Parametric analysis results for (a) NPC (b) COE.

the analysis reveals that decreasing the capital cost of the PV units, the WTs, and the batteries will decrease the NPC by 16.52 %, 6.76 %, and 18.14 %, respectively. These results provide a fruitful imprint on the economic feasibility of the designed system if any possible cost reduction of these components occurs in the future. Besides, it can be recognized that the reduction in batteries capital cost has the highest impact in reducing the NPC followed by the PV units and the WTs, respectively. Additionally, increasing the discount rate by 20 % shall reduce the project NPC to $182,718, recording a reduction ratio of 6 %.

On the contrary, reducing the discount rate to 6.6 % (a reduction by 20 %) negatively impacts the project NPC since the NPC shall increase to $208,000, recording a growth ratio of 7.2 %. The response of the project NPC against the change in project lifetime is opposite to its behaviour against the discount rate. It can be seen that increasing the project lifetime by 5 years shall increase the NPC to $216,731 (11.7 %) due to the increase of the replacement payments. Opposing, when the project lifetime decreases to 20 years, the NPC would decrease to $169,272, recording a reduction ratio of 12.75 %. In addition, increasing both the solar irradiance and wind speed by 10 % of their average annual value will save almost the same amount of money since the NPC can be reduced by 2.7 %. Besides, as expected, reducing both the solar irradiance and wind speed will increase the NPC due to the necessity for more PV units and WTs to satisfy the load requirements.

On the other side, the impact of the parametric variation on the project COE follows the same behaviour on the NPC except for the discount rate and the project lifetime. From Fig. 27(b), it can be recognized that increasing the discount rate by 20 % shall increase the COE to 0.1332$/kWh, recording an increase by 12 % while reducing the discount rate to the same ratio will reduce the COE by 10.6 % (0.106 $/kWh). These can be considered a non-constructive impact on the project's financial feasibility, but the impact on the NPC metric is more significant than the COE. Lastly, for a reference COE of 0.1191$/kWh, the project lifetime slightly impacts the COE since decreasing the lifetime to 20 years will increase the COE to 0.123$/kWh while adding 5 years more to the project lifetime will decrease the COE to 0.1169 $/kWh. Categorically, the decrease in capital costs and project lifetime and the increase of the discount rate will profitably impact the project economic behaviour, which decision-makers and investors should consider for the suitable investment preference.

In addition, in order to validate the economic performance of the optimal energy alternative, its economic values were compared to the economic outcomes of different relative case studies addressed in literature. Considering the most up-to-date and relative case studies listed in Table 1 in the introduction section and since the calculation of the NPC depends on components' capacities and project capital expenses, its significance was considerably unbalanced and varied from a region to

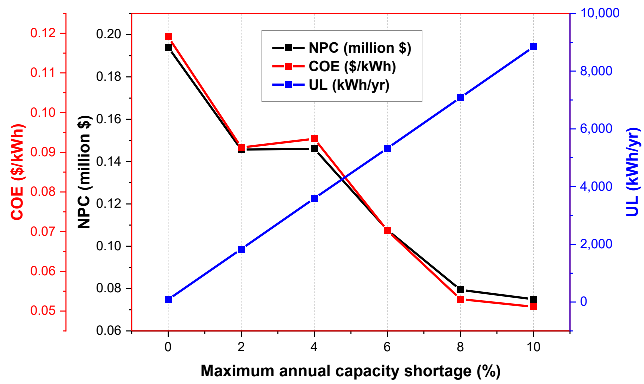

Fig. 28. Impact of CSF% variation on the economic and energy performance indices of the microgrid system.

another. Instead, the COE values can be observed as a consistent metric of energy projects cost. By evaluating the COE of different relative case-studies which lead to an average value of 0.2502 $/kWh, the COE of the proposed case-study (0.119 $/kWh) was found below the average which would extend the awareness into the economic feasibility of the proposed system.

Reliability analysis for the microgrid system

For a comprehensive investigation of the microgrid performance indicators and optimal design, reliability analysis is performed. To do so, different values for the maximum annual capacity shortage denoted as CSF are considered from 0 % to 10 % with a step of 2 % while optimizing the components' sizes. Accordingly, the results of reliability analysis are given in Fig. 28 and Table 7. As explained before, the fully reliable system (i.e., reference optimal design at 0 % CSF) necessitates the integration of 2 WTs and 200 batteries in addition to 85 kW PV array size as it requires a higher capacity of the WT to fulfill the demand completely in the cloudy periods. On the other side, the decline of CSF to 2 % gives a considerable reduction in NPC and COE by 24.9 % and 23.4 %, respectively and increase in unmet load by 1837.768 kWh/year. In that case, despite increasing the PV array size to 162 kW and the number of WTs to 3, the elimination of the batteries from the optimal system configuration results in reducing the overall system capital and operating costs; thus, NPC and COE values. This show that if the loss of power supply for that tiny period is tolerable with a fair customer satisfaction level, particularly in the agriculture and irrigation sector, the financial benefits are noticeably high. Almost, the same reductions in NPC and COE are observed at 4 % CSF, in which the water pumping system is supplied by a larger PV array size of 234 kW without wind or battery systems. The WT system is added again with the optimal design in addition to the PV when reaching 8 % CSF. This happens owing to the complementary nature of solar and wind coming into play and keeps the battery out of the system.

With the more increase in CSF up to 10 % CSF, the NPC and COE continue to drop reaching $74,951 and 0.0511$/kWh and the unmet load rise to 8840.439 kWh/year. The economic analysis of that case results in significant reductions of 61.4 % and 57.1 % NPC and COE, respectively compared to the reference design. These cost reductions are due to the reduction in the total system initial capital and operating costs because of removing the WTs and BATs, despite using a higher size for PV array of 112 against 85 kW with the reference design.

Conclusions

In this study, the techno-economic feasibility of standalone fully renewable energy systems for energizing a 240 kWh/day water pumping application in an agriculture area in Al-Sadat city, Egypt, was investigated. HOMER Pro optimizer was used for exploring the technical and economic visibility of a set of energy alternatives: (1) standalone PV/CONV (the existing system) as the base-case, (2) PV/BAT/CONV, (3) WT/BAT/CONV, (4) PV/WT/BAT/CONV (5) isolated diesel generator for fair comparisons. The parametric analysis was also implemented to expect the influence of the uncertainty of seven key input parameters on the optimal system behaviour. The simulation, optimization, and parametric analysis lead to the following findings:

- ⊗ The assessment of the energy alternatives uncovered that the integration of an 85 kW PV system, 2x10 kW WTs, 200 batteries, and 30 kW power converter rating have the best techno-economic behaviour compared to others.
- ⊗ The capacities mentioned above gave the least NPC and COE with only $194,017 and $0.119/kWh, respectively, guaranteeing the practicability of a fully sustainable energy option with a renewable fraction of 100 %.
- ⊗ The existing real system (the PV/CONV system) is considered the most economic system; however, results revealed that the unmet load is 45,073 kWh/yr which accounts for more than 50 % of the electricity consumption need of the irrigation facilities. This is attributed to the great capacity shortage fraction of this alternative (51.5 %) due to the reliance on a single renewable supply and the undersized calculation of the PV system. Consequently, this system is found impractical from a reliability point of view.
- ⊗ Besides, the existing system generates untapped surplus energy of 15.36 % that cannot be stored due to the absence of an energy storage element.
- ⊗ Meanwhile, the optimal system has a negligible capacity shortage fraction and an unmet load ratio of 0.098 %, which clarifies the proposed system's robustness and high consistency. Further, a great amount of surplus energy (61.7 %) can be exploited for further deferrable load or selling.

Table 7
Optimal microgrid design results with different reliability levels.

CSF %	Optimal design capacities				Performance indicators									
					Economic						Technical			
	PV kW	WT Qty	BAT Qty	CONV kW	NPC $	COE $/kWh	OPEX $	CAPEX $	NPC reduction %	COE reduction %	UL kWh/yr	PV Production kWh/yr	WT Production kWh/yr	BAT Throughput kWh/yr
0	85	2	200	30	194,017	0.1191	3868	122,032	0.0 %	0.0 %	87	140,509	97,384	5,815
2	162	3	–	32	145,628	0.0912	2223	104,254	24.9 %	23.4 %	1,838	268,032	146,077	–
4	234	–	–	30	146,069	0.0934	2503	99,480	24.7 %	21.6 %	3,591	386,063	–	–
6	168	–	–	30	107,691	0.0703	1849	73,289	44.5 %	41.0 %	5,329	277,923	–	–
8	67	2	–	43	79,416	0.0530	1189	57,297	59.1 %	55.5 %	7,085	110,608	97,384	–
10	112	–	–	30	74,951	0.0511	1290	50,944	61.4 %	57.1 %	8,840	185,542	97,384	–

- ⊗ Moreover, the paramedic analysis findings showed the significant reliance of system cost on the uncertainty of discount rate, project lifetime, and components' capital cost, while the uncertainty in renewable resources activity slightly impacts the system performance. Decreasing the capital cost of PV units, WTs, and batteries will decrease the NPC by 16.52 %, 6.76 %, and 18.14 %, respectively. Besides, increasing the discount rate by 20 % shall reduce the NPC by 6 % while reducing it to 6.6 % (a reduction by 20 %) hurts the NPC since the NPC shall increase by 7.2 %.
- ⊗ Finally, the findings of the microgrid reliability analysis highlight the influence of the power supply shortage and availability on the optimal microgrid configuration with the prospect of attaining more savings in the microgrid cost against a tiny and tolerable decrease in CSF%. The results show that CSF% is inversely proportional to NPC and COE values of the optimal microgrid system, where the rise of CSF from 0 % up to 10 % lead to the decline of NPC and COE by 61.4 % and 57.1 %, respectively.

CRediT authorship contribution statement

Mahmoud F. Elmorshedy: Formal analysis, Visualization, Conceptualization, Writing – original draft, Writing – review & editing. **Mohamed R. Elkadeem:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – review & editing. **Kotb M. Kotb:** Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Ibrahim B.M. Taha:** Funding acquisition, Investigation, Writing – review & editing. **Mohamed K. El-Nemr:** Investigation, Supervision, Writing – review & editing. **A.W. Kandeal:** Writing – review & editing. **Swellam W. Sharshir:** Conceptualization, Resources, Writing – original draft, Project administration. **Dhafer J. Almakhles:** Writing – review & editing, Funding acquisition. **Sherif M. Imam:** Conceptualization, Resources, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table A1 describe the specifications of the microgrid elements of photovoltaic module, wind turbine, diesel generator, battery energy storage, and power converter. The model type of each element is PV Sun330, Eocycle EO10, Generic Genset, BAT Trojan SAGM 12 75, and Generic bi-directional, respectively. In addition, Table A1 includes the technical specifications such as rating power, rating voltage, lifetime, and other specific data according to each component. On the other side, Table A2 summarizes the cost data of each element in the proposed microgrid system where all different types of costs are included such as capital cost, replacement cost, O&M cost, and fuel price., the sources of these costs are also cited in Table A2.

Table A1
Technical specifications of the microgrid elements

Component	Item	Value
PV	Rated power	0.33 kW
	Nameplate voltage	46.20 V
	Nameplate current	9.38 A
	Temperature coefficient	-0.4294
	Normal operating cell temperature	44.2 ^{o}C
	Efficiency at STC	13%
	Derating factor	85%
	Ground reflectance	0%
	Lifetime	25 years
WT	Rated power	10 kW
	Cut-in velocity	2.75 m/s
	Cut-off velocity	20 m/s
	Rated velocity	6.5 m/s
	Hub height	23 m
	Lifetime	20 years
Diesel	Fuel curve slope	0.273 L/hr/kW output
	Intercept coefficient	0.033 L/hr/kW rated
	Lifetime	15,000 hrs
	Minimum load ratio	25%
	Carbon Monoxide	16.34 g/L of fuel
	Nitrogen Oxides	15.359 g/L of fuel
	Lower heating value	43.2 MJ/kg
	Price of diesel fuel	0.429 $/L
	Carbon content	88%
	Sulfur content	0.4%
	Fuel density	820 kg/m^3
	Maintenance schedule	None
	CHP heat recovery ratio	0%

(*continued on next page*)

Table A1 (*continued*)

Component	Item	Value
Battery	Nominal voltage	12 V
	Nominal capacity	1.02 kWh
	Maximum capacity	84.8 Ah
	SOC limits	20~100%
	Roundtrip efficiency	85%
	Capacity ratio	0.614
	String size	50
	Maximum charge current	15 A
	Maximum discharge current	190 A
	Lifetime throughput	504.9 kWh
Converter	Efficiency	95%
	Relative capacity	100%
	Lifetime	15 years

Table A2
Cost data of the microgrid elements

Component	Capital cost	Replacement cost	O&M cost	Fuel price	Source
PV	400 $/kW	400 $/kW	10 $/kW/yr	–	[63]
WT	11,000 $/turbine	10,000 $/turbine	30/turbine/yr	–	[64]
DG	1000 $/kW	1000 $/kW	0.02 $/op.hr/yr	0.429$/L	[63–65]
CONV	200 $/kW	200 $/kW	0 $/kW/yr	–	[66]
BAT	300 $/battery	150 $/battery	10 $/battery/yr	–	[67]

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